

FunWake-Codex: A Frontier-Agent Replication of the Narrow-Interface Discovery Result

Anonymous Authors¹

Abstract

Prior work introduced FunWake, a benchmark that exposes LLM agents to a narrow (4-output schedule) vs. broad (write an arbitrary optimizer) interface on the same wind-farm-layout problem, and reported that Anthropic Claude Code and Google Gemini CLI both *discovered novel schedule parameterizations* given the narrow interface but *re-implemented community-standard solvers* given the broad one. We replicate that experiment with a third frontier agentic CLI, OpenAI Codex (gpt-5.5), running 3 independent agent seeds in each interface (6 runs total). [PLACEHOLDER]: Codex’s schedule-only runs reach a held-out AEP of TBD GWh (+TBD vs. baseline), structurally TBD relative to Claude/Gemini’s discoveries. Codex’s full-optimizer runs converge on TBD across seeds. The narrow-vs-broad asymmetry TBD for Codex, providing TBD evidence that the FunWake narrow interface elicits algorithmic novelty across the major frontier agentic CLIs of 2026.

1. Background

We use FunWake as defined in the companion paper. Briefly: the benchmark places N wind turbines inside a polygon to maximize annual energy production (AEP) subject to boundary and minimum-spacing constraints, using a differentiable Gaussian wake model (Bastankhah & Porté-Agel, 2014). Two interfaces are exposed to the agent:

Narrow. The agent writes `schedule_fn(step, total_steps, lr0, alpha0)` returning `(lr, alpha, beta1, beta2)`, controlling a fixed 8000-step Adam skeleton. The skeleton is identical across agents

¹Anonymous Institution, Anonymous City, Anonymous Region, Anonymous Country. Correspondence to: Anonymous Author <anon.email@domain.com>.

Preliminary work. Under review by the International Conference on Machine Learning (ICML). Do not distribute.

and submissions, so evaluation cost is independent of the schedule’s complexity.

Broad. The agent writes an arbitrary `optimize(sim, n_target, boundary, min_spacing, wd, ws, weights)` function. The agent controls everything: initialization, solver, constraint handling, and stopping.

The training cell is DEI ($N=50$, observed wind rose); the held-out cell is ROWP ($N=74$, IEA 10 MW turbine, Weibull rose). The 500-multi-start TopFarm SGD (Réthoré et al., 2014) baseline scores 5540.7 GWh on DEI and 4246.7 GWh on ROWP.

In the companion paper, Claude Code and Gemini CLI given the narrow interface produced novel schedule structures (dual Gaussian LR bumps for Claude, 8-cycle cosine restarts with cyclic Adam betas for Gemini) that beat the baseline on the held-out cell. Given the broad interface, both reimplemented SLSQP (Kraft, 1988) or wrapped `topfarm_sgd_solve`.

2. Codex Method

We add OpenAI Codex CLI v0.125 (gpt-5.5) as a third frontier agentic CLI, alongside Claude Code and Gemini CLI. Codex runs locally via the `codex exec` non-interactive command under a workspace-write sandbox. Each agent run uses the same hot-start (`results/seed_schedule.py` for narrow, `results/seed_optimizer.py` for broad), the same `agent_memory.md` cross-iteration scaffolding, the same `attempt-log + scoring` tools, and an `AGENTS.md` task spec mirroring the per-mode CLAUDE.md/GEMINI.md prompts used for Claude/Gemini.

Time budget. Schedule-only saturation in Claude/Gemini occurred near attempt 150 in the original 5 hr runs. Using the empirical formula $t_{\text{total}} = (1 + \text{think}_{\text{factor}}) \cdot N \cdot t_{\text{seed}}$ with $N=150$, $t_{\text{seed}} \approx 30$ s, and $\text{think}_{\text{factor}} \approx 1$ (empirical), we allocate 3 hr per schedule-only run and 4.5 hr per full-optimizer run.

Table 1. Codex schedule-only across 3 agent seeds. Best feasible script per run, scored on the held-out ROWP cell. Comparators are Claude/Gemini single-best from the companion paper.

RUN	TRAIN	ROWP	STRUCTURESAGENT	NARROW	BROAD
CODEX SEED 1	TBD	TBD	TBD	DISCOVERY (BUMPS)	REIMPL (SGD WRAPPER)
CODEX SEED 2	TBD	TBD	TBD	DISCOVERY (COS)	REIMPL (SLSQP)
CODEX SEED 3	TBD	TBD	TBD	TBD	TBD
CODEX MEAN	TBD	TBD	—	—	—
±STD	TBD	TBD	—	—	—
CLAUDE (COMPANION BEST)	5555.7	4271.5	DUAL-BUMPS	[matching Claude), or invent a third approach? Does the broad-interface modal choice flip across seeds?]	
GEMINI (COMPANION BEST)	5565.1	4269.3	COS-RESTART		
500-START BASELINE	5540.7	4246.7	—		

Table 2. Codex full-optimizer across 3 agent seeds. “Modality” classifies the high-level algorithmic approach.

RUN	TRAIN	ROWP	MODALITY
CODEX SEED 1	TBD	TBD	TBD
CODEX SEED 2	TBD	TBD	TBD
CODEX SEED 3	TBD	TBD	TBD
GEMINI (BEST OF 5)	5557.1	4272.7	SLSQP
CLAUDE RUN 1	5559.5	4264.9	SGD-WRAPPER
500-START BASELINE	5540.7	4246.7	—

Seeds. We launch 3 independent agent runs per interface (6 runs total). Codex CLI single-account concurrency caps the chain to one run at a time; total wall-clock ≈ 22.5 hr.

Scoring. For each run we extract the highest-training-AEP feasible script and score it on the held-out ROWP cell. We additionally classify each script’s structural motif via regex matching against a fixed taxonomy (Gaussian bumps, cosine restarts, cyclic betas, alpha-squeeze for narrow; SLSQP-wrapper, SGD-wrapper, custom-Adam, GA, CMA-ES for broad).

3. Results

3.1. Schedule-only ($n = 3$)

[PLACEHOLDER: 1–2 paragraphs describing the Codex schedule-only results from Table 1. Key questions to address: (a) does Codex beat the baseline on the held-out cell? (b) is the variance across 3 seeds comparable to Claude/Gemini’s init-variance? (c) does Codex rediscover Claude’s bump motif, Gemini’s cosine-restart motif, or a third independent structure?]

3.2. Full-optimizer ($n = 3$)

[PLACEHOLDER: 1–2 paragraphs describing the Codex full-optimizer results. Key questions: did Codex rediscover SLSQP (matching Gemini), wrap topfarm_sgd_solve

Table 3. Per-agent narrow-vs-broad outcome, including Codex. Discovery = novel schedule structure; Reimplementation = community-standard solver wrapper.

3.3. Narrow vs broad summary

[PLACEHOLDER: short interpretation of Table 3.]

4. Discussion

[PLACEHOLDER: 1 paragraph on whether Codex strengthens or weakens the companion paper’s central claim. Three possible outcomes:

- All three agents discover-narrow / reimplement-broad: the claim strengthens substantially, becomes "frontier-agent-invariant".
- Codex breaks the pattern (e.g., reimplements under narrow): the claim narrows to "Claude-and-Gemini-specific" and we should investigate why Codex behaves differently.
- Mixed: Codex matches on one mode but not the other; report which specific transition fails.

]

Limitations. $n=3$ agent seeds per cell; Codex agent-run variance not measured beyond the 3 included runs. Single training farm, single held-out polygon (inherited from the companion paper). The structural classification is regex-based; a human review of all six scripts is provided in the supplement to verify motifs.

5. Conclusion

[PLACEHOLDER: 2–3 sentences on whether Codex extends the FunWake narrow-interface discovery result to a third frontier provider, and what the 2026 frontier-agent landscape implies for using narrow interfaces as a general-purpose algorithm-discovery elicitation mechanism.]

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal

consequences of our work, none which we feel must be specifically highlighted here.

References

- Bastankhah, M. and Porté-Agel, F. A new analytical model for wind-turbine wakes. *Renewable Energy*, 70:116–123, 2014.
- Kraft, D. A software package for sequential quadratic programming. Technical Report DFVLR-FB 88-28, DFVLR (Deutsche Forschungs- und Versuchsanstalt für Luft- und Raumfahrt), 1988.
- Réthoré, P.-E., Fuglsang, P., Larsen, G. C., Buhl, T., Larsen, T. J., and Madsen, H. A. TOPFARM: Multi-fidelity optimization of wind farms. *Wind Energy*, 17(12):1797–1816, 2014. doi: 10.1002/we.1667.